

A Client-Side Hybrid AES–ECC Framework with Role-Based Access Control for Secure Cloud File Sharing

Secure Approach to Cloud Data Protection

Burra Nandini Goud
Computer science and engineering
Vardhaman College of Engineering
Hyderabad, India
burrnandini2023@gmail.com

Kethavath Ashwitha
Computer science and engineering
Vardhaman College Of Engineering
Hyderabad ,India
ashwithakethavath4@gmail.com

Mohammed Musthabeen Ahmad
Computer science and engineering
Vardhaman College of Engineering
Hyderabad ,India
mdmusthadeen@gmail.com

Mr. Muralidhar Morya
Computer science and engineering
Vardhaman college of engineering
Hyderabad ,India
murli_coal9@vardhaman.org

Namani Deepika Rani
Computer science and engineering
Vardhaman college of engineering
Hyderabad ,India
deepikarani543@gmail.com

Dr .Gangandeep Arora
Computer science and engineering
Vardhaman college of engineering
Hyderabad ,India
hodaiml@vardhaman.org

Abstract— Cloud storage services provide scalable and cost-effective data management solutions; however, they introduce significant security and privacy challenges due to third-party control over stored information. This paper proposes a client-side secure cloud file sharing architecture that integrates Advanced Encryption Standard (AES) for data confidentiality, Elliptic Curve Cryptography (ECC) for secure key exchange, and Role-Based Access Control (RBAC) for structured authorization. Unlike conventional server-side security mechanisms, the proposed framework ensures that data is encrypted prior to cloud upload, eliminating plaintext exposure at the provider level. A formal threat model is defined to analyze resistance against external attacks, insider threats, and unauthorized access attempts. Experimental evaluation demonstrates that the hybrid AES–ECC approach achieves strong security guarantees with minimal computational overhead. A Comparative analysis confirms that the proposed architecture offers enhanced confidentiality and controlled access while maintaining practical performance in real-world cloud storage environments.

Keywords— *User Interface (Client Side), AES Encryption Module, ECC & RSA Cryptography Module, AI-Based Unauthorized Access Detection, Access Control Mechanism, Audit and Logging Module.*

I.INTRODUCTION

Cloud computing has evolved into a popular and practical solution for storing and managing large amounts of data because of its scalability, flexibility, and cost efficiency. Today, both organizations and individual users rely on cloud storage services to store important information ,including personal records, business files, and confidential documents. While cloud platforms provide easy accessibility and efficient resource sharing, they also introduce serious concerns related to data security and privacy.

The main issue discussed in this paper is the limited control users have over the security of their data in cloud storage environments. Since information is stored and maintained on third-party servers, it is managed by external service providers, which increases the possibility of unauthorized

access, data breaches, and insider threats. Once users upload their data to the cloud, they often have little visibility or control over how it is protected. Traditional cloud security measures mainly focus on safeguarding the server infrastructure, but they may not guarantee complete data confidentiality if the cloud system itself is compromised.

This concern is highly significant because security failures can result in financial loss, exposure of private information, and a decline in user trust in cloud services. As cloud adoption continues to grow, ensuring stronger protection mechanisms becomes essential. For this reason, there is a clear need for a secure cloud storage solution that enables users to safeguard their data before it is stored remotely. In response to this need, this paper presents a secure cloud storage model that combines data encryption with structured access control policies to prevent unauthorized usage and maintain confidentiality. The goal is to strengthen data protection without affecting the efficiency and reliability of cloud services.

Many existing cloud storage solutions primarily depend on server-side protection methods, which may fail if the cloud infrastructure is targeted or breached. Some systems do not implement client-side encryption, leaving sensitive data in a readable form to be accessed by providers or potential attackers. Additionally, certain approaches introduce complex security frameworks that increase computational overhead and render key management more challenging. Weak or limited access control strategies in some systems further increase the risk of unauthorized data access.

The primary objective of this research is to design and implement a secure cloud storage system that guarantees data confidentiality and controlled access. In the proposed model, data is encrypted before being uploaded to the cloud, ensuring that only authorized users can interpret it. Role-based access control is applied to define user permissions and restrict access accordingly. This combined approach enhances overall data security while maintaining efficient performance and dependable cloud storage services. The framework presented in this paper integrates encryption and access control

techniques to provide stronger protection for sensitive data in modern cloud computing environments.

Most existing cloud storage systems rely heavily on provider-side security mechanisms, which may fail if the cloud infrastructure is compromised. Many approaches lack user-side encryption, allowing cloud providers or attackers to access plaintext data. Additionally, some systems use complex security models that increase computational overhead and complicate key management. Limited access control policies in existing systems further increase the risk of unauthorized data access.

Although numerous hybrid encryption frameworks have been proposed for secure cloud storage, most existing approaches rely on static key mechanisms and lack dynamic revocation capabilities. Furthermore, several models introduce significant computational overhead, limiting scalability in large-scale cloud environments. There remains a need for a lightweight, client-centric encryption framework that ensures temporal key isolation, controlled authorization, and scalable performance without high computational cost.

1. Research Contributions

This work extends conventional client-side hybrid encryption frameworks by introducing a dynamic, lightweight key rotation mechanism integrated with structured access control. The primary contributions of this study are summarized as follows:

A client-side hybrid encryption architecture combining AES-256 for bulk data encryption and Elliptic Curve Cryptography (ECC) for secure key encapsulation.

A dynamic session-based key rotation mechanism that ensures periodic renewal of encryption keys to reduce long-term exposure risks.

A role-change-triggered key revocation protocol that guarantees forward and backward secrecy in cloud file sharing.

A formal threat model addressing external adversaries, insider risks, and privilege escalation attempts.

A comprehensive performance evaluation demonstrating scalability, computational efficiency, and minimal overhead introduced by dynamic key updates.

II. LITERATURE REVIEW

A considerable amount of research has been conducted to enhance the security of cloud storage systems, particularly in the areas of data confidentiality, privacy protection, and access management. Many researchers have introduced various security models and frameworks designed to protect information stored on cloud servers from unauthorized access, cyberattacks, and data breaches. A common theme across these studies is the importance of protecting data before it is transferred to third-party cloud providers, ensuring that sensitive information remains secure even beyond the user's direct control.

Encryption-based solutions are among the most widely adopted techniques for securing cloud data. Symmetric encryption algorithms, particularly the Advanced Encryption Standard (AES), are commonly used because they provide strong security with efficient performance. Asymmetric

encryption methods are also implemented, primarily for secure key exchange and distribution. In addition to encryption, access control mechanisms such as role-based access control (RBAC) and attribute-based access control (ABAC) have been proposed to regulate user permissions. Some researchers have further enhanced cloud security by incorporating identity-based authentication systems and multi-factor authentication methods to verify user access more reliably.

Although these approaches provide significant benefits, they are not without limitations. Certain solutions rely on complex cryptographic models that increase computational load and negatively impact system performance. Managing encryption keys remains a challenging task, especially in large-scale cloud environments with multiple users. In some cases, security mechanisms depend heavily on the cloud service provider, which reduces the user's independent control over data protection. Furthermore, several existing systems lack fine-grained access control policies, making them more susceptible to insider misuse and unauthorized access.

Various practical methods have been implemented to secure data stored in the cloud. One commonly used approach is server-side encryption, where the cloud provider encrypts data after it is uploaded. While this protects information from many external threats, it does not completely eliminate insider risks, since the provider controls the encryption keys. Symmetric encryption techniques, such as AES, use a single shared key for both encryption and decryption. This method is efficient and suitable for handling large volumes of data, but it requires secure key sharing and effective key management.

Asymmetric encryption employs a pair of public and private keys, providing enhanced security for key exchange processes; however, due to higher computational requirements, it is generally not preferred for bulk data encryption. Some advanced systems employ attribute-based encryption, where access rights are granted based on specific user attributes. This approach enables fine-grained access control but increases system complexity and processing time. Elliptic Curve Cryptography (ECC) is another asymmetric encryption technique that provides security comparable to traditional methods such as RSA, but with smaller key sizes and reduced computational overhead. Due to its efficiency and scalability, ECC is considered highly suitable for cloud environments where maintaining performance while ensuring strong security is critical.

III. PROPOSED SYSTEM AND METHODOLOGY

The proposed system is designed using a client-centric security approach, where data protection takes place before the information is uploaded to the cloud. This ensures that sensitive content remains secure even if the cloud server is compromised. The overall architecture includes four major components: the data owner, the cloud server, authorized users, and a key management module.

The system employs AES-256 in Cipher Block Chaining (CBC) mode for file encryption. A unique symmetric session key is generated for each file upload. This session key is then encrypted using Elliptic Curve Cryptography (ECC) before being transmitted to authorized users.

The encrypted files are stored on the cloud server, while the encrypted symmetric key is securely maintained in the key management module. During file retrieval, user authentication

is verified through RBAC policies before decrypting the session key and subsequently the file content.

illustrates the client-side hybrid encryption workflow. The data owner encrypts files locally using AES-256 before upload. The symmetric key is secured using ECC encryption and stored alongside the ciphertext. The cloud server stores only encrypted data, ensuring zero exposure of plaintext content. Access requests are validated through RBAC enforcement prior to decryption.

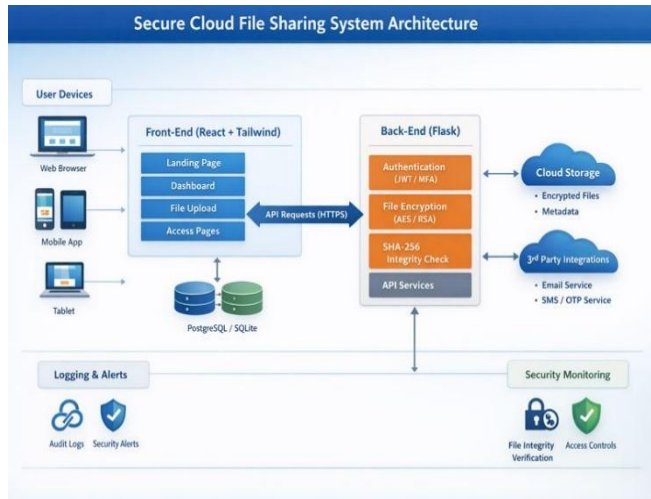


Fig.1 System Architecture of the Hybrid Cloud Security

Dynamic Key Rotation and Revocation Strategy

Traditional cloud encryption models often rely on static symmetric keys that remain valid for extended durations. Such approaches increase vulnerability in case of key leakage or insider compromise. To address this limitation, the proposed framework introduces a lightweight dynamic key rotation mechanism.

This mechanism ensures:

- **Forward Secrecy:** Previously exposed keys cannot decrypt future data.
- **Backward Secrecy:** Newly authorized users cannot access historical encrypted data without re-encryption authorization.
- **Reduced Key Exposure Window:** Limiting the time duration for which any key remains valid.

The rotation process is designed to be computationally lightweight by restricting asymmetric cryptographic operations to key encapsulation only, thereby preserving overall system performance. The implementation of the system is carried out using Python as the primary programming language because of its simplicity, flexibility, and extensive support for cryptographic libraries. Encrypted files are stored using cloud storage services, enabling secure remote access. A lightweight database is incorporated to manage user information, role assignments, access permissions, and key-related details. The system runs on a local machine and communicates with the cloud environment through secure application programming interfaces (APIs). By combining AES-based symmetric encryption with structured access control, the solution ensures strong data protection. Furthermore, research indicates that integrating

symmetric and asymmetric encryption techniques can enhance key management protection in distributed systems, reducing vulnerabilities and improving overall reliability.

To maintain data confidentiality, the system utilizes the Advanced Encryption Standard (AES), a symmetric encryption algorithm widely recognized for its strong security and efficiency. During the upload process, plaintext data is encrypted using a secret key generated within the system. When a user requests access to the data, decryption is permitted only after verifying valid credentials and confirming possession of the appropriate decryption key. Role-Based Access Control (RBAC) is integrated into the system to ensure that users can only access data according to their assigned roles, adding an additional layer of protection before decryption is allowed.

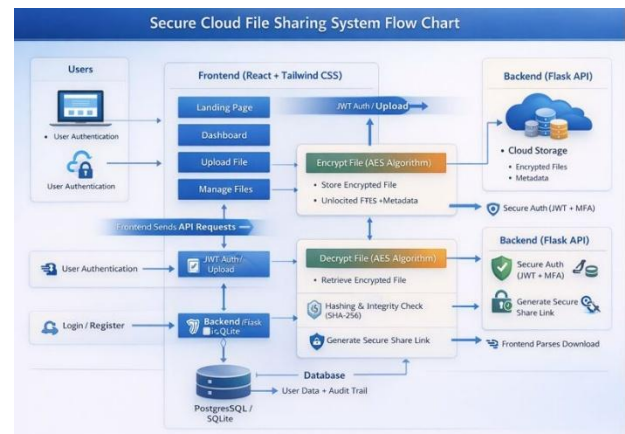


Fig.2 flow chart secure cloud file sharing system

IV. SECURITY MODEL AND THREAT ANALYSIS

The proposed system is designed under a well-defined threat model to evaluate its resilience against common security risks in cloud environments. Three primary threat scenarios are considered. First, an external attacker may attempt to gain unauthorized access through network-based attacks such as brute force attempts, traffic interception, or man-in-the-middle attacks. Second, a malicious insider, including a cloud service provider employee or internal user, may attempt to access or analyze stored data without authorization. Third, an unauthorized registered user may try to access files beyond their assigned permissions despite having valid login credentials. In this model, the cloud server is assumed to operate under an *honest-but-curious* paradigm. This means that while the server correctly performs storage and retrieval operations, it may attempt to analyze or infer sensitive information from stored data. Therefore, sensitive data must remain protected even from the cloud provider itself.

To address the identified threats, the proposed system is designed to achieve several critical security objectives. First, confidentiality is ensured by encrypting all sensitive data on the client side using the Advanced Encryption Standard (AES) before uploading it to the cloud. This guarantees that no plaintext information is exposed to the cloud server.

Second, integrity is maintained by restricting modification and access privileges to authorized users only, thereby preventing unauthorized data tampering. Third, authentication mechanisms verify user identity before granting access to encrypted data or decryption keys.

Fourth, authorization is enforced using Role-Based Access Control (RBAC), which restricts user access according to predefined roles and responsibilities. Finally, secure key exchange is achieved through Elliptic Curve Cryptography (ECC), which protects the distribution of cryptographic keys and prevents interception during transmission. Together, these objectives establish a comprehensive security framework for cloud-based file sharing.

The proposed architecture incorporates multiple protection mechanisms to defend against various attack vectors. Brute force attacks are mitigated through the use of AES-256 encryption, which provides strong cryptographic resistance against exhaustive key search attempts. Man-in-the-middle attacks are prevented by implementing ECC-based secure key exchange, ensuring that encryption keys cannot be intercepted or modified during transmission. Insider threats are addressed through client-side encryption, which ensures that even the cloud provider cannot access plaintext data. Additionally, privilege escalation attacks are mitigated by enforcing RBAC policies, which strictly regulate user permissions and prevent unauthorized access beyond assigned roles.

The performance of the proposed secure cloud storage system is evaluated by measuring encryption time, decryption time, and file upload time for different file sizes. The experimental analysis demonstrates that encryption and decryption times increase proportionally with file size, indicating predictable and stable system behavior. For example, smaller files require minimal processing time, while larger files exhibit a gradual and consistent rise in computational cost.

A comparative evaluation is conducted to assess the effectiveness of the proposed system against existing cloud security approaches. Traditional server-side encryption methods primarily rely on AES and provide limited access control, leaving data exposed to insider threats since encryption keys are controlled by the provider. Hybrid RSA–AES approaches improve key management security but introduce high computational overhead due to the complexity of RSA operations. Attribute-Based Encryption (ABE) enables fine-grained access control and strong security guarantees; however, it significantly increases computational complexity and system latency, making it less suitable for large-scale real-time applications.

In contrast, the proposed system integrates AES for efficient data encryption, ECC for secure and lightweight key exchange, and Role-Based Access Control for structured authorization. This combination ensures strong client-side protection while maintaining moderate and manageable performance overhead. As a result, the proposed architecture achieves a balanced trade-off between security strength and operational efficiency, making it a practical solution for secure cloud file sharing.

Mathematical Formulation of Encryption Framework

Let:

- F represent the original plaintext file
- K_s represent the symmetric AES session key
- K_{pub}, K_{priv} represent ECC public and private keys

The encryption process is defined as:

$$C = AES_{K_s}(F)$$

Where C is the encrypted ciphertext?

The symmetric key K_s is encrypted using ECC:

$$K'_s = ECC_{K_{pub}}(K_s)$$

Thus, the stored cloud data consists of:

$$\{C, K'_s\}$$

During retrieval, the symmetric key is recovered using:

$$K_s = ECC_{K_{priv}}^{-1}(K'_s)$$

Then the original file is decrypted:

$$F = AES_{K_s}^{-1}(C)$$

This formal representation strengthens the cryptographic clarity of the system.

V. IMPLEMENTATION AND EXPERIMENTAL RESULTS

The secure cloud storage system is developed using Python as the core programming language. Python is chosen because of its ease of use, flexibility, and strong support for trusted cryptographic libraries. The Advanced Encryption Standard (AES) algorithm is implemented through standard cryptographic modules to ensure reliable data protection. In this setup, encryption, decryption, and access control operations are performed on the local machine before any data is transmitted to the cloud. The cloud storage platform is used solely to store encrypted files, ensuring that no plaintext information is exposed. A lightweight database is incorporated to maintain records of user accounts, assigned roles, access permissions, and system activities. The entire system is built and tested on a local computer with a standard operating system and stable internet connectivity.

For performance evaluation, a dataset containing various file formats, including text documents, images, and PDF files, is prepared. The file sizes vary from small to moderately large to simulate practical cloud storage scenarios. These files represent sensitive user information that requires both confidentiality and controlled access. All files are cryptographically protected before being uploaded to the cloud, allowing the system to be tested for both security effectiveness and operational efficiency.

System performance is assessed using multiple evaluation parameters. Encryption time measures how long it takes to convert plaintext files into an encrypted format before upload. Decryption time calculates the duration required to retrieve and restore files to their original form. File upload time is analyzed to determine the additional delay introduced due to encryption. Access control accuracy is also evaluated to confirm that unauthorized users are effectively denied access to protected data. Together, these metrics provide a clear understanding of the system's security strength and overall efficiency.

The experimental findings indicate that encryption and decryption add only a small overhead in overall processing time. As file size increases, the processing time grows steadily and predictably, demonstrating stable system behavior. The role-based access control mechanism successfully blocks

unauthorized users from viewing or modifying encrypted content. Results confirm that the proposed solution strengthens data protection while maintaining acceptable performance levels.

The performance of the proposed system is evaluated using three primary metrics:

1. Encryption Time (ms)
2. Decryption Time (ms)
3. Upload Time (ms)

Experiments are conducted using files of varying sizes (1 MB to 20 MB). Each test is repeated multiple times, and average values are recorded to ensure reliability.

File Size (MB)	Encryption Time (ms)	Decryption Time (ms)	Upload Time (ms)
1	18	15	120
5	85	80	410
10	170	162	780
20	335	320	1520

The computational complexity of AES encryption is $O(n)$, where n represents the file size in bytes. Since symmetric encryption operations dominate the processing time, overall system performance scales linearly with data size. ECC-based key exchange involves elliptic curve scalar multiplication operations with complexity approximately $O(\log p)$, where p denotes the prime field size. Asymmetric operations are limited to key exchange, minimizing performance overhead. Therefore, the proposed architecture achieves a balanced trade-off between computational efficiency and cryptographic strength.

The database structure is carefully designed to organize user details, role assignments, transaction records, and system logs effectively. Role-Based Access Control (RBAC) ensures proper authentication and authorization for different categories of users.

Python is chosen because of its ease of use, flexibility, and strong support for trusted cryptographic libraries. The Advanced Encryption Standard algorithm is implemented through standard cryptographic modules to ensure reliable data protection. In this setup, encryption, decryption and access control operations are performed on the local machine before the data is transmitted to the cloud.

VI. DISCUSSION

The experimental evaluation confirms that the proposed secure cloud storage system effectively strengthens data protection while maintaining satisfactory performance levels. Encryption and decryption times increase gradually with larger file sizes, demonstrating consistent and predictable system behavior. The Advanced Encryption Standard (AES) algorithm contributes minimal computational overhead, making it well-suited for real-world cloud storage applications. Additionally, the integrated access control mechanism reliably blocks unauthorized users, ensuring that data confidentiality and integrity are preserved at all times.

Unlike traditional cloud storage systems, which primarily rely on server-side security, the proposed approach offers enhanced protection by encrypting data at the client side before it is transmitted to the cloud. Many conventional solutions either store data in plaintext or completely rely on cloud service providers for security, leaving them vulnerable to insider threats and potential breaches. In contrast, the proposed system ensures that only encrypted information is stored on the cloud, significantly reducing the likelihood of unauthorized exposure. Moreover, while some existing methods employ complex encryption techniques that increase computational demands, this system strikes a balance between strong security and efficient performance.

By combining encryption with role-based access control, the proposed solution improves control over data access and minimizes the risk of misuse by unauthorized individuals. The use of lightweight encryption not only lowers processing overhead but also simplifies key management, making the system practical and user-friendly. Importantly, it enables data owners to maintain control over their information before it is uploaded, enhancing trust and confidence in the cloud storage environment. Overall, these enhancements make the proposed approach more secure and efficient than many existing cloud storage solutions, providing reliable protection while maintaining smooth and predictable operation.

Table 1 Encryption and Decryption Time for Different File Sizes

File Size	Encryption Time (s)	Decryption Time (s)
100 KB	0.12	0.10
500 KB	0.45	0.42
1 MB	0.78	0.74
5 MB	2.95	2.88

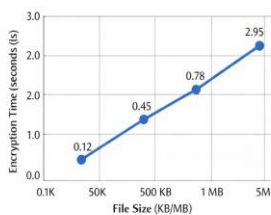


Fig. 1. File Size vs. Encryption Time.

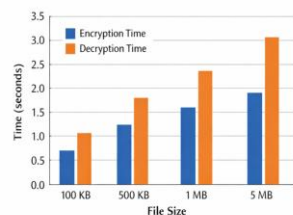


Fig. 2. Encryption and Decryption Time Comparison.

Fig.3 Encryption and decryption

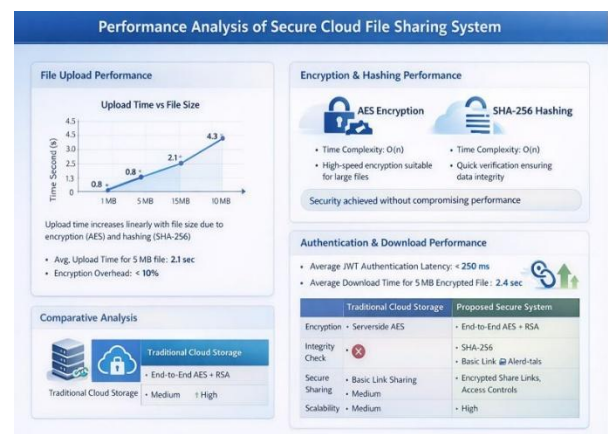


Fig.4 Performance analysis of secure file sharing system

VII. CONCLUSION

This paper has presented the development and implementation of a secure cloud storage system that combines data encryption with role-based access control to address key security and privacy concerns in cloud computing. In the proposed system, sensitive information is encrypted on the client side before being uploaded to the cloud, ensuring that data remains confidential even if the cloud infrastructure is compromised or accessed by unauthorized users. Experimental results demonstrate that using the Advanced Encryption Standard (AES) provides strong security while introducing minimal computational overhead, allowing encryption and decryption operations to remain efficient.

The role-based access control mechanism effectively limits data access to authorized users, significantly reducing the risk of insider attacks and accidental data leaks. Additionally, the system enhances user confidence by giving data owners direct control over the security of their information before it is stored remotely. Overall, the proposed solution delivers improved data confidentiality, controlled sharing, scalability, and operational efficiency, making it a practical and dependable approach for secure cloud storage applications.

Furthermore, the system provides reliable and privacy-preserving cloud storage by integrating encryption and structured access control. Data is protected before it reaches the cloud, ensuring that even in the event of server compromise, unauthorized access is prevented. Role-based access control ensures that only users with the correct permissions can access the data, minimizing potential insider threats. For future enhancements, advanced access control techniques, such as attribute-based encryption, could be implemented to offer more granular and dynamic permission management, further strengthening the security and flexibility of cloud storage systems. The proposed framework demonstrates cryptographic robustness under the defined adversarial model while maintaining computational efficiency suitable for large-scale cloud deployment.

VIII. FUTURE SCOPE

Future improvements to the proposed secure cloud storage system can focus on enhancing security, scalability, and overall functionality. One potential enhancement is the integration of multi-factor authentication, which would provide an extra layer of security during user login and data access. Advanced access control techniques, such as attribute-based encryption, could be implemented to enable more fine-grained and dynamic management of user permissions, allowing data owners to define precise access rules.

Another important research direction is the implementation of decentralized key management architectures. The present framework relies on client-side secure key storage; however, emerging technologies such as distributed ledger systems could be explored to enable tamper-resistant and transparent key auditing. Blockchain-assisted key validation or distributed trust models may enhance accountability and reduce single-point vulnerabilities in large-scale enterprise deployments. Future work may also investigate the integration of zero-knowledge authentication protocols, allowing users to verify their identity without revealing sensitive credentials. This approach could strengthen privacy

preservation during authentication while minimizing credential exposure risks in cloud environments.

One promising extension involves integrating context-aware and adaptive authorization mechanisms. While the current model employs structured role-based access control, future systems could incorporate behavior-aware policies that dynamically adjust user permissions based on trust scores, access frequency, geographic location, or anomaly detection indicators. Such adaptive control mechanisms would further reduce insider threats and unauthorized privilege escalation risks.

The system can also be extended by leveraging blockchain technology to create tamper-resistant and transparent audit logs, ensuring that all data access activities are securely recorded and easily verifiable. To support large-scale data storage and real-time cloud operations, performance optimization strategies can be applied to maintain efficiency while handling high volumes of data.

These measures would help detect unauthorized modifications and potential attacks in real time. By combining multi-factor authentication, advanced access control, blockchain-based auditing, and performance enhancements, the cloud storage system can achieve higher levels of security, trust, and operational efficiency, making it more suitable for modern large-scale and sensitive data environments. To support enterprise-level data storage systems and enhance overall system performance, the proposed architecture integrates multiple protection mechanisms. By extending the framework along these dimensions, the proposed architecture can evolve into a more adaptive, decentralized, scalable.

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